Leaving Presentation

James Bray – Summer Placement

TOMORROW.
DONE BETTER.

About me

Name: James Bray

Role: Assistant Project Engineer (Electrification Team)

Duration: 3-month placement (leaving tomorrow)

Education: Aerospace Engineer MEng (3rd year)

Professional Interests: CAD/CAM, design, simulation

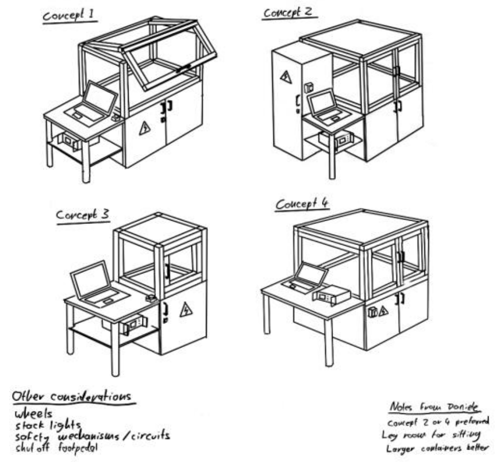
Personal Interests: Running, nature, my beardie Ozzie

Project 1 – Partial Discharge Test Rig

Task: Design and build an enclosure for safely executing 20kV partial discharge tests on motor coils

Challenges: Annual leave season and supplier issues leading to project delays

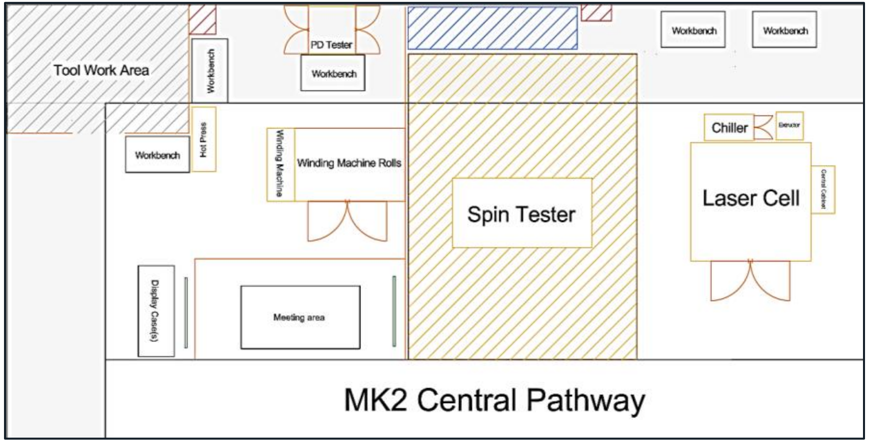
Outcome: A speed build this week, but hopefully a completed test cell for MK2 before I leave

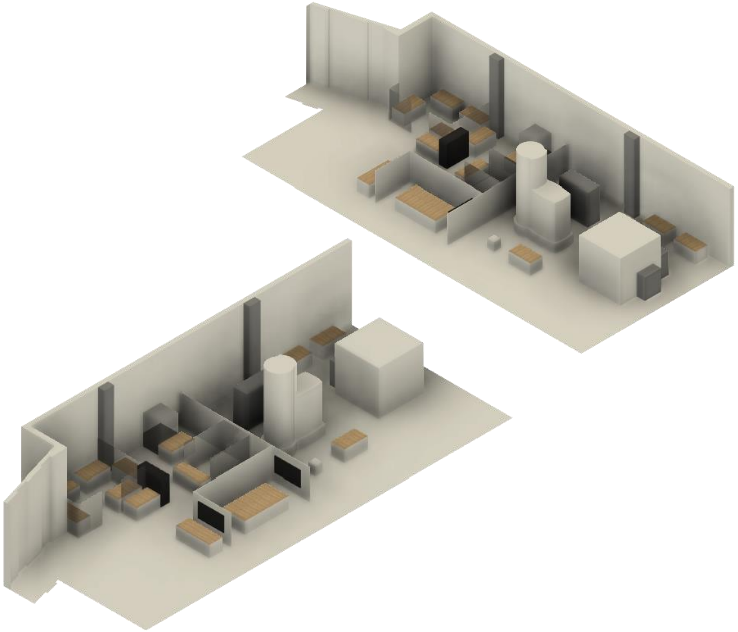


Project 2 – MK2 Electrification Redesign

Task: Create a layout plan for the electrification area in MK2, incorporating a new meeting area

Outcome: A new layout plan with separated display, meeting and work areas



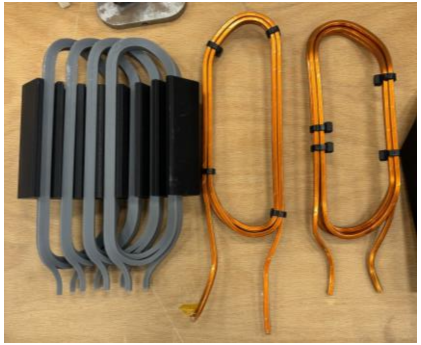


Project 3 – Winding Guide for Axial Flux Motor Coils

Task: Adjust the existing CAD to wind coils in a different configuration, then print the new part

Challenges: A few iterations was needed due to minor design oversights

Outcome: A 3D printed part ready to be used in the winding machine

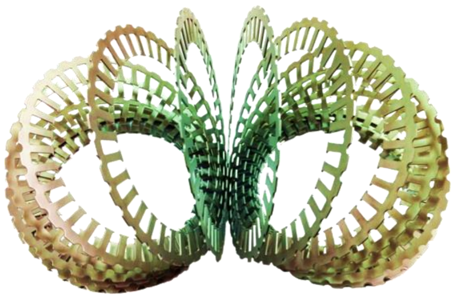


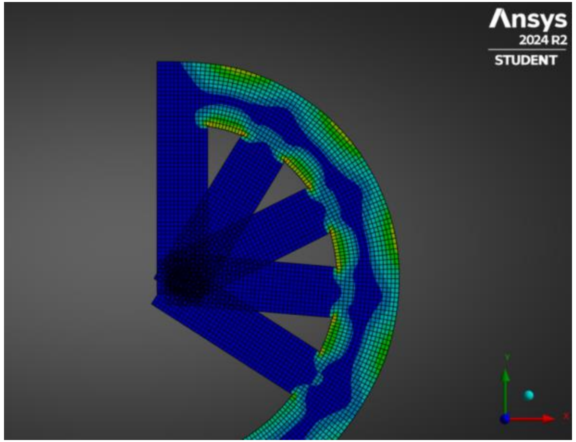
Project 4 – Slinky Stator FEA

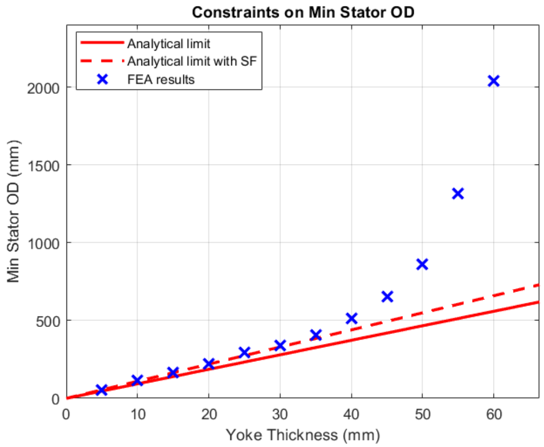
Task: Run non-linear FEA on stator geometries to determine feasibility of manufacture

Challenges: I hadn’t used FEA before; inaccuracies of plastic deformation in FEA

Outcome: Results were as expected to see, providing limits to the geometry that can be manufactured





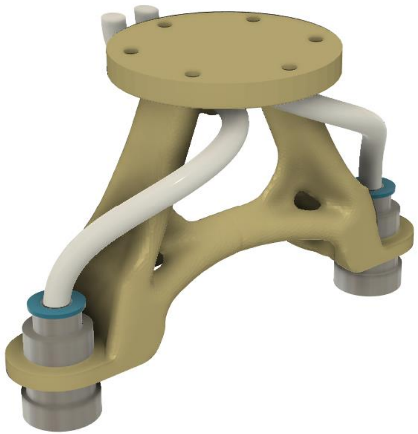


Project 5 – Optimised Pneumatic End-Effector

Task: Design and print an optimised end effector for the Igus ReBeL robotic arm

Challenges: I needed to learn how to perform design optimisation in Ansys

Outcome: An optimised 3D printed part which included pipe routing

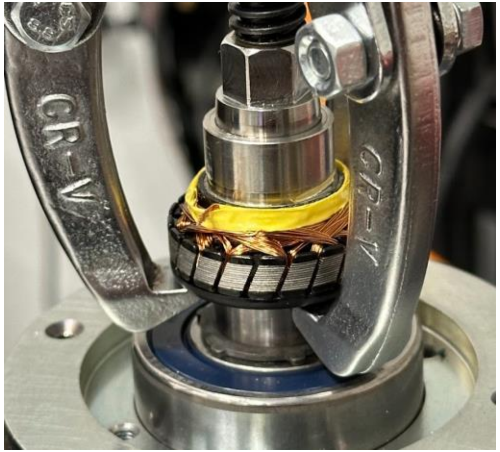


Project 6 – Motor Disassembly

Task: Determine how much of the Siemens 1FK7 Servo disassembly process could be automated

Challenges: Design problem with the resolver, bearing pullers couldn’t be used

Outcome: Managed to remove the resolver… as expected…



Statistic Breakdown

- Distance cycled to and from work – **679.4 km**
- Number of projects contributed to – **6**
- Burritos meal prepped – **38**
- IT issues raised – **26**
- Average daily coffee consumption – **2.76 cups**
- CAD software crashes – **583**
- Souvenirs crafted – **1**

Thank you for the opportunity to be a part of this team.

It's been a pleasure to learn and contribute during my time here.

Feel free to add my LinkedIn and stay connected :)

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